Virtual Reality Simulation for Fire Safety Emergency Training

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Problem Statement: Gaps in Fire Safety Training

1. Traditional methods of safety training often lack realism, are treated as procedural formalities, and are not retained well.

2. Large-scale drills in institutions can be difficult to organise as well as manage.

3. Safely recreating realistic fire scenarios for training purposes without exposing participants to real danger remains a challenge.

Project Objectives: Bridging the Gap with VR

Design and Implementation

Design and implementation of an interactive virtual reality application for fire safety emergency training.

Guided Training Flow

A structured training sequence guiding users through methods of fire handling and suppression.

Scenario Development

Development of controlled indoor fire scenarios to support practical training.

Functional Scope

Provide a safe and repeatable environment for practicing emergency response procedures.

Design Principles of the VR Training System

IMMERSION

Controlled virtual environment fully occupying the user's perception.

+

INTERACTION

Enabling natural user actions and responses within the virtual fire scenario.

+

SENSORY FEEDBACK

Visual and auditory responses that communicate feedback during interaction

PRESENCE & REALISM

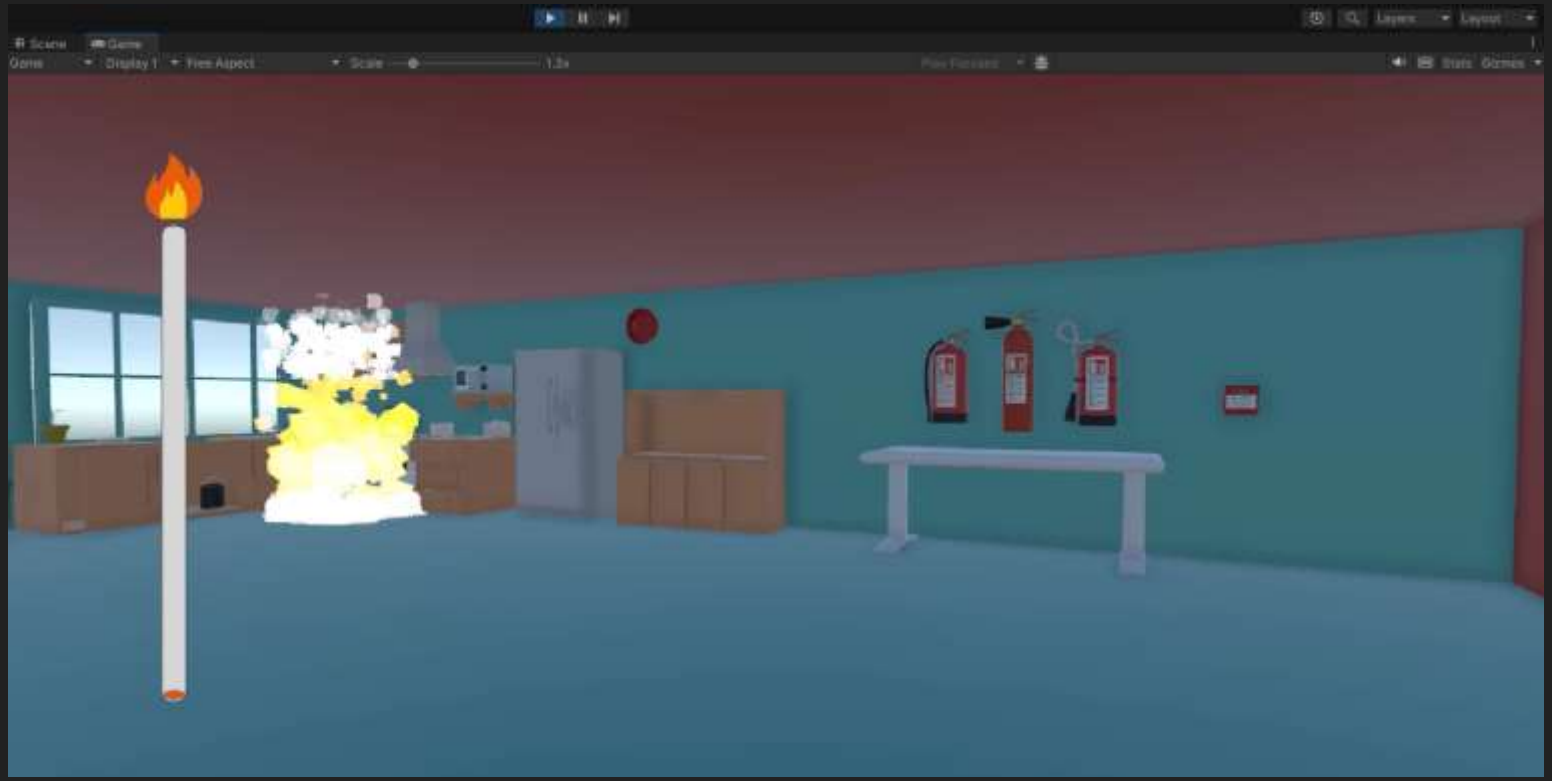
Technologies Used for VR Fire Safety Training

Primary platform for building the virtual environments and running the VR simulations.

Blender was used to model and create 3D assets for the training environments and interactive objects.

Event logic handled using C# scripts to manage user interactions and system responses within the scenario.

VR Training System Walkthrough



VR Training Scenario Logic

1. Scenario Initialization
A fire scenario starts in a controlled indoor environment, prompting the user to respond.

2. Alarm Activation
The user activates the fire alarm, triggering audio cues and advancing the scenario.

3. Extinguisher Selection
The user selects an extinguisher, with guided visual hints if no action is taken.

4. Fire Suppression
Fire is suppressed by aiming and spraying the extinguisher continuously at the source.

5. Scenario Completion
Fire, smoke, and audio effects are stopped, ending the training sequence.

Conclusions

Designed and implemented a VR simulation guiding users through the core steps of fire handling.

Shifting training from passive observation to hands-on practice.

VR can be used to rehearse safety procedures in situations where real-world training would be risky or impractical.

Future Development

Extending the simulation beyond single rooms to larger connected spaces.

Designing fires that change over time, spread unpredictably, and react to user decisions.

Supporting performance tracking and teamwork scenarios involving multiple users.

Enhancing realism through haptic feedback and mixed reality features.

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